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B.Tech. Degree VIII Semester Supplementary Examination August 2018

ME 1801 COMPRESSIBLE FLUID FLOW (2012 Scheme)

Time : 3 Hours

Maximum Marks : 100

(Approved gas tables are permitted)

PART A

(Answer *ALL* questions)

(8 × 5 = 40)

- I. (a) Sketch adiabatic ellipse for steady and compressible flow. Indicate different regimes of flow.
- (b) Explain geometric choking.
- (c) The state of a gas ($\gamma = 1.3$, $R = 0.469 \text{ kJ/kg.K}$) upstream of a normal shock wave is given by the following data $M_x = 2.5$, $P_x = 2 \text{ bar}$, $T_x = 275 \text{ K}$. Calculate Mach number, temperature and velocity of flow downstream of the shock.
- (d) Plot $\theta v/s \beta$ curve for constant M and write down the conditions for getting
 - (i) Strong attached shock wave
 - (ii) Weak attached shock wave
 - (iii) Detached shock wave
- (e) Explain thermal choking.
- (f) Explain friction choking.
- (g) How does Prandtl Pitot static tube simultaneously measure the static and stagnation temperatures?
- (h) Explain temperature recovery factor and discuss its significance.

PART B

(4 × 15 = 60)

- II. (a) Derive an expression for velocity of sound waves in a compressible fluid. (9)
 - (b) The pressure, temperature and Mach number at the entry of a flow passage are 2.45 bar, 26.5°C and 1.4 respectively. If the exit Mach number is 2.5, determine for adiabatic flow of a perfect gas ($\gamma = 1.4$, $R = 0.287 \text{ kJ/kgK}$)
 - (i) Stagnation temperature
 - (ii) Temperature and velocity of gas at exit
 - (iii) The flow rate per square metre of the inlet cross section
- OR**
- III. (a) Derive an expression for ratio of impulse function F and F^* in isentropic flow as a function of Mach numbers and ratio of specific heats. (7)
 - (b) Air flowing in a duct has a velocity 300 m/s, pressure 1.0 bar and temperature 290 K. Taking $\gamma = 1.4$ and $R = 287 \text{ J/kgK}$, determine
 - (i) Stagnation pressure and temperature
 - (ii) Velocity of sound in the dynamic and stagnation conditions
 - (iii) Stagnation pressure assuming constant density.

(P.T.O.)

- IV. Derive Prandtl-Meyer relation for normal shock wave and obtain expressions for Mach number downstream and static pressure ratio across the shock. (15)

OR

- V. Derive Rankine-Hugoniot equation and Prandtl's equation for oblique shock wave. (15)

- VI. Derive an expression for the term $4fL_{\max} / D$ as a function of initial Mach number and ratio of specific heats. (15)

OR

- VII. (a) What is Rayleigh flow? Under what conditions is the assumptions for Rayleigh flow, not valid in heat exchanger. (7)

- (b) A combustion chamber in a gas turbine plant receives air 350 K, 0.55 bar and 75 m/s. The air fuel ratio is 29 and the calorific value of fuel is 41.87 MJ/kg for the gas. Determine, (8)

- (i) The initial and final Mach numbers.
- (ii) Final pressure, temperature and velocity of gas.
- (iii) Percent stagnation pressure loss in the combustion chamber.
- (iv) The maximum stagnation temperature attainable.

- VIII. (a) Explain three different optical instrument employed in compressible flow to study the density variation in flow field. (7)

- (b) Explain the principle of Schlieren techniques of flow visualizations. (8)

OR

- IX. Explain in detail about wind tunnels, its different types and uses with neat figures.. (15)
